

**HANOI UNIVERSITY OF SCIENCE
FACULTY OF MATHEMATICS, MECHANICS AND
INFORMATICS**

LAB REPORT

COMPUTER VISION (MAT3562E 2)

**LAB 10: HISTOGRAM OF ORIENTED GRADIENTS (HOG)
AND ITS APPLICATIONS**

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1 Overview and Learning Objectives

This report completes all core requirements for Lab 10:

1. Manual HOG implementation from scratch using NumPy.
2. Comparison against `scikit-image` HOG.
3. Two practical applications: image classification and classical object detection.
4. Error analysis with quantitative evidence.

The implementation is in: `lab10_BuiQuangChien_23001837.ipynb`.

2 Dataset and Experimental Setup

2.1 Data used

- Traffic sign image (and fallback camera image) for manual HOG visualization.
- `sklearn.datasets.load_digits()` for quantitative classification and detector experiments.

2.2 HOG configuration

The same HOG configuration is used across manual and library pipelines:

- Orientation bins: 9.
- Cell size: 8×8 .
- Block size: 2×2 cells.
- Block normalization: L2-Hys.

2.3 Core equations

Given Sobel gradients G_x, G_y :

$$M(x, y) = \sqrt{G_x(x, y)^2 + G_y(x, y)^2}, \quad (1)$$

$$\theta(x, y) = \left(\text{atan2}(G_y, G_x) \cdot \frac{180}{\pi} + 180 \right) \bmod 180. \quad (2)$$

For each block vector v , L2-Hys normalization is:

$$\hat{v} = \frac{v}{\sqrt{\|v\|_2^2 + \epsilon^2}}, \quad (3)$$

$$\tilde{v} = \min(\hat{v}, 0.2), \quad (4)$$

$$v_{\text{L2-Hys}} = \frac{\tilde{v}}{\sqrt{\|\tilde{v}\|_2^2 + \epsilon^2}}. \quad (5)$$

3 Manual HOG Implementation

The manual pipeline includes:

1. Preprocessing and resize.
2. Sobel gradient computation.
3. Orientation quantization and magnitude voting.
4. Histogram per cell.
5. Block normalization and feature concatenation.
6. Visualization of dominant local orientations.

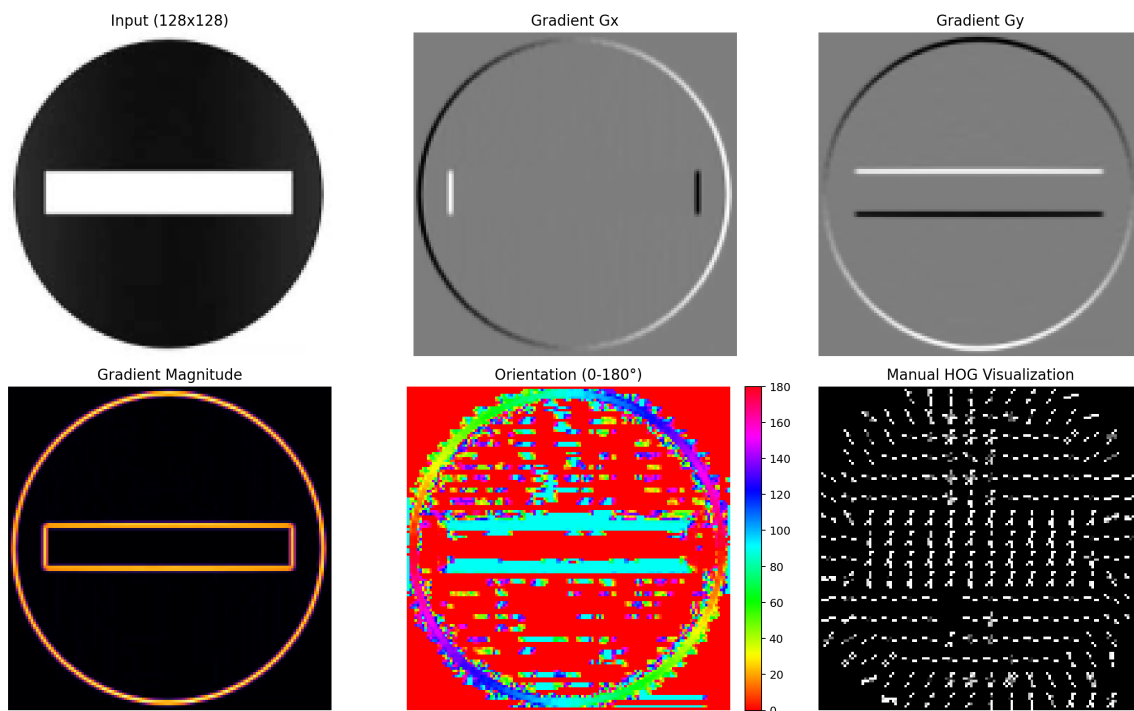


Figure 1: Manual HOG pipeline visualization (input, gradients, magnitude, orientation, HOG view).

On the 128×128 sample image, the manual feature vector length is 8100, matching library output.

4 Manual vs Library Comparison

4.1 Feature dimensions and speed

Table 1: Manual HOG vs `scikit-image` HOG

Metric	Manual	<code>scikit-image</code>
Feature length (sample image)	8100	8100
Single-image extraction time (ms)	10.80	4.19
Batch extraction time (ms/image)	0.790	0.190

The library implementation is faster, while the manual implementation provides full algorithmic transparency.

5 Application 1: Image Classification

5.1 Pipeline

1. Upscale digits to 32×32 .
2. Extract HOG features (manual or library).
3. Train classifiers: LinearSVC and Logistic Regression.
4. Evaluate using accuracy and confusion matrix.

5.2 Model comparison

Table 2: Classification results from `outputs/classification_model_comparison.csv`

Feature + Classifier	Accuracy	Train time (s)	Feature dim
Manual + LogisticRegression	0.9889	0.5568	324
Skimage + LogisticRegression	0.9889	0.2579	324
Manual + LinearSVC	0.9822	4.0223	324
Skimage + LinearSVC	0.9756	2.9581	324

Best classification configuration: Manual HOG + Logistic Regression with accuracy 0.9889.

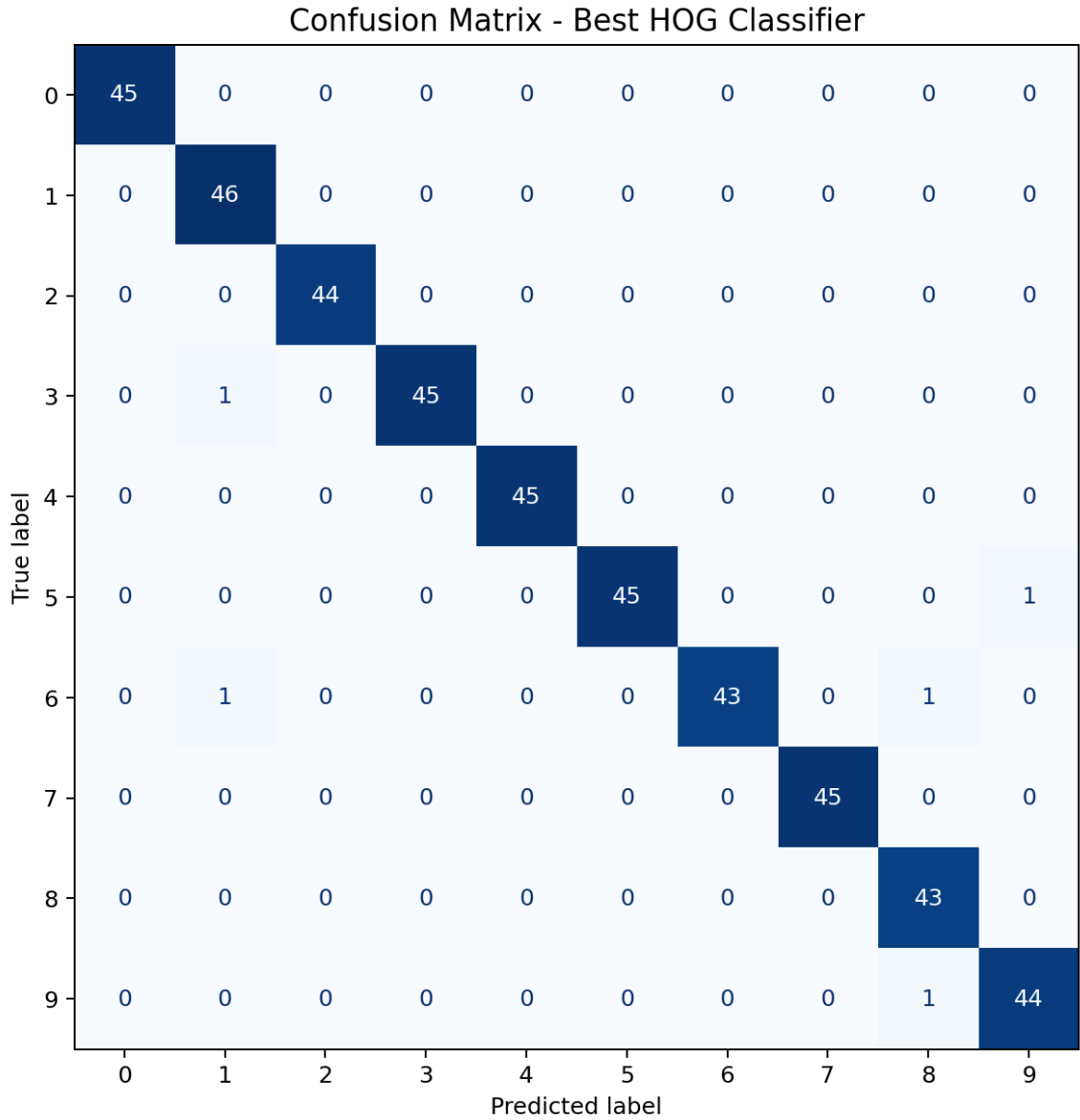


Figure 2: Confusion matrix of the best classifier.

6 Application 2: Classical HOG Object Detector

6.1 Detector design

The detector follows the classical pipeline:

1. Image pyramid for multi-scale search.
2. Sliding window scan over each scale.
3. HOG extraction and linear SVM scoring.
4. Non-Maximum Suppression (NMS) with IoU thresholding.

6.2 Detector classifier quality

On the binary detector training split (digit 0 vs non-0):

- Accuracy: 0.9900
- Precision: 0.9706
- Recall: 1.0000
- F1-score: 0.9851
- PR-AUC: 0.9991

6.3 Threshold tuning to reduce false positives

Grid search over detector thresholds on 10 synthetic scenes gives best configuration:

- Score threshold: 0.75
- NMS IoU threshold: 0.15
- Aggregated precision: 0.3548
- Aggregated recall: 0.8800
- Aggregated F1: 0.5057

Table 3: Single-scene comparison: baseline vs tuned thresholds

Setting	TP	FP	FN	Precision	Recall	F1
Baseline (0.35, 0.25)	5	15	0	0.250	1.000	0.400
Tuned (0.75, 0.15)	5	7	0	0.417	1.000	0.588

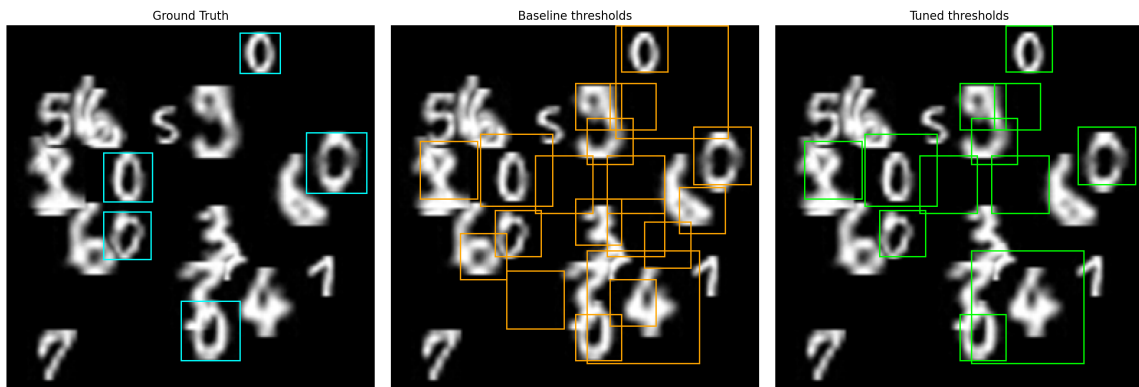


Figure 3: Detector output: baseline thresholds vs tuned thresholds.

7 Error Analysis

7.1 Classification errors

Per-class metrics remain high for all classes (lowest F1 around 0.9773). Top confusion pairs are sparse and each appears once:

- $3 \rightarrow 1$
- $5 \rightarrow 9$
- $6 \rightarrow 1$
- $6 \rightarrow 8$
- $9 \rightarrow 8$

This indicates strong class separation from HOG features on the digits task.

7.2 Detection errors

From `outputs/detection_summary_metrics.csv`:

Table 4: Aggregated detection error summary after tuning

Metric	Value
Total TP	44
Total FP (near-miss)	6
Total FP (background)	74
Total FN	6
Precision	0.3548
Recall	0.8800
F1	0.5057
Mean IoU on TP	0.6669
FN small / medium / large	3 / 3 / 0

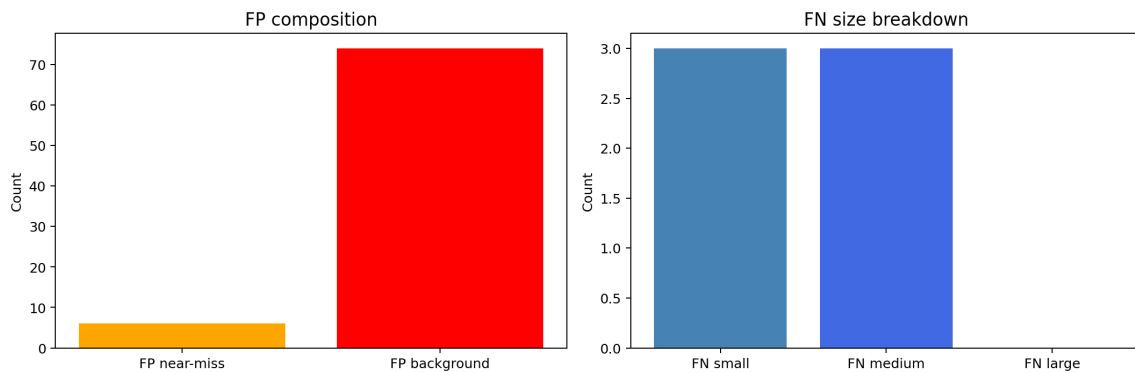


Figure 4: False-positive composition and false-negative size distribution.

Main observations:

- Background false positives dominate near-miss false positives.
- False negatives mostly appear in small and medium objects; no large-object misses.
- This is consistent with HOG sensitivity to local edge patterns and sliding-window discretization.

8 Conclusion

Manual HOG implementation clarified every algorithmic stage and matched the library feature length. The library version is computationally faster. For classification, HOG + linear models achieved very high accuracy. For classical detection, tuning score and NMS thresholds reduced duplicate detections and improved precision on the inspected scene while keeping recall high.

9 Reproducibility Artifacts

Generated files under lab-10/outputs:

- PNG: manual_hog_pipeline.png, classification_confusion_matrix.png, detector_baseline.png, detector_error_breakdown.png
- CSV: classification_model_comparison.csv, detector_threshold_tuning.csv, classification_per_class_metrics.csv, classification_top_confusions.csv, detection_scene_level_breakdown.csv, detection_summary_metrics.csv